Anthony Christiana

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Education

The George Washington University *August 2022 – May 2027*
Ph.D. in Mathematics (expected May 2027)

- M.A. in Mathematics, 2024
- Advisor: Józef H. Przytycki
- Dissertation: *Braid Algebras*

Hamilton College *August 2018 – May 2022*
B.A. in Mathematics and Music, magna cum laude

- Honors in Mathematics
- Phi Beta Kappa

Awards & Honors

George Washington University *Washington, DC*

Columbian Distinguished Fellowship 2026–2027
Funded by the Mary Hopkins Shepard Endowed Graduate Fellowship for Science

Graduate Student Teaching Prize in Mathematics 2026

Marvin Green Prize 2026

James H. Taylor Graduate Mathematics Prize 2025

Hamilton College *Clinton, NY*

Edward Huntington Memorial Mathematics Prize 2022

Madeleine Wild Bristol Scholarship 2022

Carter Family Prize Scholarship 2022

Research Interests

My research interests include knot theory, quandles, diagrammatic algebras, skein modules, homology of quasigroups, domination in graphs, and solutions to the Yang–Baxter equation.

Publications & Manuscripts

Dates for unpublished papers refer to the initial preprint year.

Published and Accepted

Using Fibonacci numbers and Chebyshev polynomials to express Fox coloring groups and Alexander–Burau–Fox modules of diagrams of wheel graphs 2026

Anthony Christiana, Huizheng Guo, and Józef H. Przytycki

Contemporary Mathematics 834 (2026), 109–129.

doi:10.1090/conm/834/16687 arXiv:2310.19979

Low Dimensional Homology of the Yang–Baxter Operators Yielding the HOMFLYPT Polynomial 2025

Anthony Christiana, Ben Clingenpeel, Huizheng Guo, Jinseok Oh, Józef H. Przytycki, Xiao Wang, and Hongdae Yun

Accepted for publication in *Homology, Homotopy and Applications*.

arXiv:2502.20659

Chebyshev polynomials and Gram determinants from the Möbius band 2025

Anthony Christiana, Dionne Ibarra, and Gabriel Montoya-Vega

Accepted for publication in *Contemporary Mathematics*.

arXiv:2504.16439

Psyquandle coloring quivers 2023

Jose Ceniceros, Anthony Christiana, and Sam Nelson

Journal of Knot Theory and Its Ramifications 32(11) (2023), 2350073.

doi:10.1142/S0218216523500736

Preprints and Submitted Manuscripts

On The Secure Domination of Mycielskian Graphs 2026

Kiran R. Bhutani, Anthony Christiana, and Peter Ulrickson

Preprint; submitted to *Ars Combinatoria*.

arXiv:2603.17897

Fundamentals of cubic skein modules 2025

Rhea Palak-Bakshi, Anthony Christiana, Huizheng Guo, Dionne Ibarra, Louis H. Kauffman, Gabriel Montoya-Vega, Sujoy Mukherjee, Józef H. Przytycki, and Xiao Wang

Preprint; submitted to *Advances in Mathematics*.

arXiv:2511.10959

In Search of Homology for Quasigroups of Bol–Moufang Type 2025

Anthony Christiana, Ben Clingenpeel, Huizheng Guo, Jinseok Oh, Józef H. Przytycki, and Anna Zamojska-Dzienio

Conditionally accepted for conference proceedings; revised version submitted.

arXiv:2508.21268

Manuscripts in Preparation

Gram Determinants of Type Mb and Representations of Periodic Temperley–Lieb Algebras

Anthony Christiana, Dionne Ibarra, and Gabriel Montoya-Vega

In preparation.

Teaching

Instructor of Record

MATH 1221: Calculus with Precalculus 2, George Washington University — Summer 2026

MATH 1007: Mathematics and Politics, George Washington University — Summer 2026

MATH 111: Calculus for Social & Life Sciences, Catholic University of America — Summer 2025

MATH 1007: Mathematics and Politics, George Washington University — Summer 2024

MATH 1007: Mathematics and Politics, George Washington University — Summer 2023

Teaching Assistant, George Washington University

MATH 1009: Mathematical Ideas — Spring 2026

MATH 1009: Mathematical Ideas — Fall 2025

MATH 1009: Mathematical Ideas — Spring 2025

MATH 1009: Mathematical Ideas — Fall 2024

MATH 1232: Single-Variable Calculus II — Spring 2024

MATH 1231: Single-Variable Calculus I — Fall 2023

MATH 1232: Single-Variable Calculus II — Spring 2023

MATH 1231: Single-Variable Calculus I — Fall 2022

Selected Guest Teaching

Georgetown University: Graph Theory — Spring 2026
Taught a week-long guest lecture series.

Catholic University of America: Graph Theory, Advanced Graph Theory, Combinatorics — 2025, 2026
Taught a week-long guest lecture series in each of Graph Theory and Advanced Graph Theory; also gave guest lectures in Combinatorics.

Additional Teaching and Tutoring

Upward Bound Math Instructor, George Washington University — Summers 2023–2025
Designed and taught summer courses in graph theory and topology for students ages 14–18; developed a graph theory poster project connecting mathematics to students' interests.

Mathematics Tutor, Hamilton College — 2021–2022
Provided drop-in and individual mathematics tutoring.

Teaching Assistant, Hamilton College — Fall 2020

Conference Organization

Organizer, AMS Special Session on Knots and Braids, I — October 3–4, 2026
AMS Fall Eastern Sectional Meeting, George Washington University

Knots in Washington Organizer — Fall 2022 – Present
Coordinate speaker travel, travel funding, and event logistics; manage the conference website and contribute to grant applications for the semiannual research conference series.

Presentations and Talks

Undergraduate Workshops

Knot Invariants and Notions of Equivalence
Catholic University Math Club, invited workshop

Research and Expository Talks

- **Möbius Connections and Periodic Temperley–Lieb Representations** — November 7–8, 2026
  AMS Fall Western Sectional Meeting, Arizona State University, Tempe, AZ; Special Session on Inverse Problems (forthcoming)

- **Periodic Temperley–Lieb Algebras and Gram Determinants on the Möbius Band** — October 3–4, 2026
  AMS Fall Eastern Sectional Meeting, George Washington University; Special Session on Categorification, Low-Dimensional Topology, and Representation Theory
- **Music Theory Mod n** — April 24, 2026
  GW Association for Women in Mathematics Lecture and Panel, Washington, DC
- **Gram Determinants of Type Mb & Representations of the Temperley–Lieb Algebra** (poster) — April 15, 2026
  CCAS Research Showcase, George Washington University, Washington, DC
- **Homology of Bol–Moufang Quasigroups** — April 10, 2026
  New York Combinatorics Seminar, invited talk (virtual)
- **Gram Determinants of Type Mb** — March 8, 2026
  AMS Spring Western Sectional Meeting, Boise State University, Boise, ID; invited talk, Special Session on Algebraic Structures in Knot Theory and 3-Dimensional Topology
- **Invariance Results for Bol–Moufang Homology** — December 7, 2025
  Knots in Washington 52, Washington, DC
- **Bol–Moufang Quasigroups Homology** — October 26, 2025
  AMS Virtual Fall Eastern Sectional Meeting; invited talk, Special Session on Quantum Supergroups in Low-Dimensional Topology
- **Homology of Quasigroups of Bol–Moufang Type** — October 18, 2025
  AMS Fall Central Sectional Meeting, St. Louis, MO; invited talk, Special Session on Inverse Problems
- **Invariance of Bol–Moufang Homology** — October 8, 2025
  Catholic University of America Mathematics Colloquium, invited talk
- **More Relations in Cubic Skein Modules** — May 4, 2025
  AMS Spring Western Sectional Meeting, California Polytechnic State University, San Luis Obispo, CA; invited talk, Special Session on Knot Theory and Quantum Topology
- **Third Homology of the Yang–Baxter Operator** — March 28, 2025
  50th Annual New York State Regional Graduate Mathematics Conference, Syracuse, NY
- **Cubic Skein Modules of Rational and Pretzel Links** — April 26, 2024
  Distinguished Graduate Student Talk, Knots in Washington $49\frac{63}{64}$
- **Two-Bridge Algorithm for Cubic Skein Modules** — April 6, 2024
  AMS Spring Eastern Sectional Meeting, Howard University; Special Session on Skein Modules in Low-Dimensional Topology
- **Fox Coloring Group and Alexander–Burau–Fox Module of Wheel Graphs** — May 26, 2023
  Graduate Student Conference in Algebra, Geometry, and Topology, Temple University
- **Relations in Cubic Skein Modules** — May 6, 2023
  AMS Spring Western Sectional Meeting, Fresno State University; Contributed Paper Session II